LEVEL 1 DATA FLOW DIAGRAM (DFD) SPECIFICATION

System: BuildSync — A Web-Based Construction Tracking and Client Management System
Decomposition Baseline: Context Diagram (Level 0 DFD)

This specification decomposes central process 0.0 into six (6) discrete, balanced Level 1 processes. All incoming and outgoing data flows from External Entities (Client, Staff, Administrator) are 100% mathematically balanced against the Level 0 Context Diagram.

1. SYSTEM ENTITIES & PERSISTENT DATA STORES

- **[ENTITY] CLIENT:** Prospective/active clients submitting inquiries, lot surveys, viewing quotations & progress.
- **[ENTITY] STAFF:** Architects, engineers & draftsmen handling CAD plans, takeoff BOQ, and site inspections.
- **[ENTITY] ADMINISTRATOR:** Firm executives approving contracts, reviewing inquiries, assigning roles, and monitoring LGU clearance.

- **[DATA STORE D1] User Accounts & Authentication Store:** `(public.users, public.otp_codes)`
- **[DATA STORE D2] Project Inquiries & Site Feasibility Store:** `(public.client_briefs)`
- **[DATA STORE D3] Consultation & Meeting Schedules Store:** `(public.client_briefs, meeting_history)`
- **[DATA STORE D4] Blueprints, BOQ & Quotations Store:** `(public.quotations, cloud_blueprints)`
- **[DATA STORE D5] Construction Tracking & Compliance Store:** `(public.site_projects, site_milestones, photo_logs, delay_events)`
- **[DATA STORE D6] Project Showcase & Portfolio Store:** `(public.projects)`

2. DECOMPOSED LEVEL 1 PROCESSES (1.0 TO 3.0)

PROCESS 1.0: USER AUTHENTICATION & ACCOUNT MANAGEMENT

Handles multi-role credential verification, email OTP validation, face liveness KYC, and account status management.

INCOMING FLOWS: Client: Reg Details, OTP Code, Login | Staff: Login | Admin: Login, Client Account Data

OUTGOING FLOWS: Client: Reg Status, Verification Status, Auth Status | Staff: Auth Status | Admin: Auth Status, Client Account Info

PERSISTENCE: D1: User Accounts & Authentication Store

PROCESS 2.0: PROJECT INQUIRY & SITE FEASIBILITY PROCESSING

Processes client inquiry submissions, lot GPS coordinates, zoning constraints, budget ranges, and architect reviews.

INCOMING FLOWS: Client: Inquiry Details, Site Details, Requirements & Budget | Admin: Inquiry Review & Status Update

OUTGOING FLOWS: Client: Inquiry Submission Status, Site Feasibility Report | Admin: Updated Inquiry Pipeline Data

PERSISTENCE: D2: Project Inquiries & Site Feasibility Store

PROCESS 3.0: CONSULTATION & MEETING SCHEDULING

Orchestrates face-to-face and native WebRTC video appointments, calendar coordination, and meeting room codes.

INCOMING FLOWS: Client: Consultation Booking Details | Admin: Meeting & Consultation Details

OUTGOING FLOWS: Client: Booking Confirmation and Schedule | Admin: Consultation Schedule Confirmation

PERSISTENCE: D3: Consultation & Meeting Schedules Store

DECOMPOSED LEVEL 1 PROCESSES (4.0 TO 6.0)

PROCESS 4.0: DESIGN BLUEPRINT, COSTING & QUOTATION MANAGEMENT

Computes material quantities, BOQ drafts, cost estimations, and architectural technical blueprint updates.

INCOMING FLOWS: Staff: Budget & Quotation Data, Takeoff Data, Blueprint Updates | Admin: Budget & Quotation Data

OUTGOING FLOWS: Client: Preliminary Cost Estimation | Staff: Schematics, Itemized BOQ, Cost Estimates | Admin: Quotation Records

PERSISTENCE: D4: Blueprints, BOQ & Quotations Store

PROCESS 5.0: CONSTRUCTION TRACKING, SITE PROGRESS & COMPLIANCE MANAGEMENT

Coordinates site milestones, inspection photo logs, on-site weather monitoring, and LGU permits & clearance approvals.

INCOMING FLOWS: Staff: Project Data, LGU Clearances, Site Logs | Admin: Permit Data, Site Progress Updates

OUTGOING FLOWS: Staff: Submission Status, LGU Status View, Project Info | Admin: LGU Records, Site Logs & Progress Reports

PERSISTENCE: D5: Construction Tracking & Compliance Store

PROCESS 6.0: PROJECT SHOWCASE & PUBLIC CATALOG MANAGEMENT

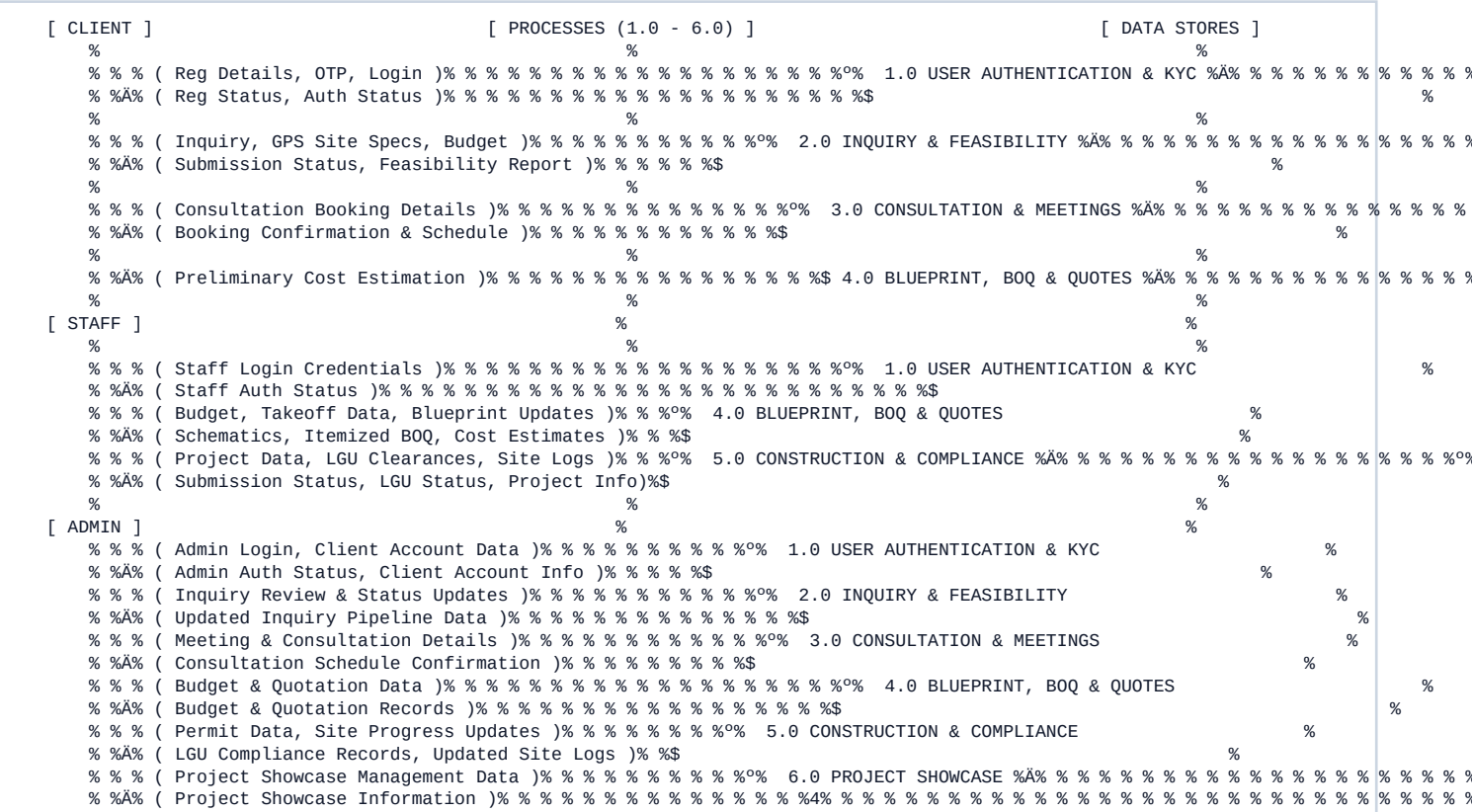
Manages public display of completed projects, architectural gallery media, and portfolio achievements.

INCOMING FLOWS: Admin: Project Showcase Management Data

OUTGOING FLOWS: Admin: Project Showcase Information

PERSISTENCE: D6: Project Showcase & Portfolio Store

3. LEVEL 1 ARCHITECTURAL DATA FLOW TOPOLOGY (DIAGRAM)



4. LEVEL 0 VS LEVEL 1 DATA FLOW BALANCING VERIFICATION MATRIX

DATA FLOW LABEL (FROM LEVEL 0 CONTEXT)	LEVEL 0 BOUNDARY	LEVEL 1 DECOMPOSED PATH	STATUS
Account Registration Details	Client -> 0.0	Client -> P1.0	BALANCED
Email Verification Code	Client -> 0.0	Client -> P1.0	BALANCED
Login Credentials (Client)	Client -> 0.0	Client -> P1.0	BALANCED
Account Registration Status	0.0 -> Client	P1.0 -> Client	BALANCED
Verification Status	0.0 -> Client	P1.0 -> Client	BALANCED
Authentication Status (Client)	0.0 -> Client	P1.0 -> Client	BALANCED
Project Inquiry Details	Client -> 0.0	Client -> P2.0	BALANCED
Project Site Details	Client -> 0.0	Client -> P2.0	BALANCED
Project Requirements and Budget	Client -> 0.0	Client -> P2.0	BALANCED
Inquiry Submission Status	0.0 -> Client	P2.0 -> Client	BALANCED
Site Feasibility Report	0.0 -> Client	P2.0 -> Client	BALANCED
Preliminary Cost Estimation	0.0 -> Client	P4.0 -> Client	BALANCED
Consultation Booking Details	Client -> 0.0	Client -> P3.0	BALANCED
Booking Confirmation and Schedule	0.0 -> Client	P3.0 -> Client	BALANCED
Staff Login Credentials	Staff -> 0.0	Staff -> P1.0	BALANCED
Staff Authentication Status	0.0 -> Staff	P1.0 -> Staff	BALANCED
Blueprint and Technical Plan Updates	Staff -> 0.0	Staff -> P4.0	BALANCED
Material and Quantity Takeoff Data	Staff -> 0.0	Staff -> P4.0	BALANCED
Budget and Quotation Data (Staff)	Staff -> 0.0	Staff -> P4.0	BALANCED
Interactive Design Schematics	0.0 -> Staff	P4.0 -> Staff	BALANCED
Itemized BOQ Draft	0.0 -> Staff	P4.0 -> Staff	BALANCED
Cost Estimates & Quotation Records	0.0 -> Staff	P4.0 -> Staff	BALANCED
Project Management Data	Staff -> 0.0	Staff -> P5.0	BALANCED
LGU Clearance Submissions	Staff -> 0.0	Staff -> P5.0	BALANCED
Site Photos and Progress Logs	Staff -> 0.0	Staff -> P5.0	BALANCED
Submission Confirmation and Status	0.0 -> Staff	P5.0 -> Staff	BALANCED
LGU Compliance Status View	0.0 -> Staff	P5.0 -> Staff	BALANCED
Project Information and Status	0.0 -> Staff	P5.0 -> Staff	BALANCED
Administrator Login Credentials	Admin -> 0.0	Admin -> P1.0	BALANCED
Client Account Management Data	Admin -> 0.0	Admin -> P1.0	BALANCED
Admin Authentication Status	0.0 -> Admin	P1.0 -> Admin	BALANCED
Client Account Information	0.0 -> Admin	P1.0 -> Admin	BALANCED
Inquiry Review and Status Update	Admin -> 0.0	Admin -> P2.0	BALANCED
Updated Inquiry Pipeline Data	0.0 -> Admin	P2.0 -> Admin	BALANCED
Meeting and Consultation Details	Admin -> 0.0	Admin -> P3.0	BALANCED
Consultation Schedule Confirmation	0.0 -> Admin	P3.0 -> Admin	BALANCED
Budget and Quotation Data (Admin)	Admin -> 0.0	Admin -> P4.0	BALANCED
Budget and Quotation Records	0.0 -> Admin	P4.0 -> Admin	BALANCED
Document & Permit Management Data	Admin -> 0.0	Admin -> P5.0	BALANCED
Project & Site Progress Updates	Admin -> 0.0	Admin -> P5.0	BALANCED
LGU Compliance Records	0.0 -> Admin	P5.0 -> Admin	BALANCED
Updated Site Logs & Progress Reports	0.0 -> Admin	P5.0 -> Admin	BALANCED
Project Showcase Management Data	Admin -> 0.0	Admin -> P6.0	BALANCED
Project Showcase Information	0.0 -> Admin	P6.0 -> Admin	BALANCED

Audit Verdict: All forty-four (44) data flows are strictly conserved with zero dangling sinks or orphaned sources.